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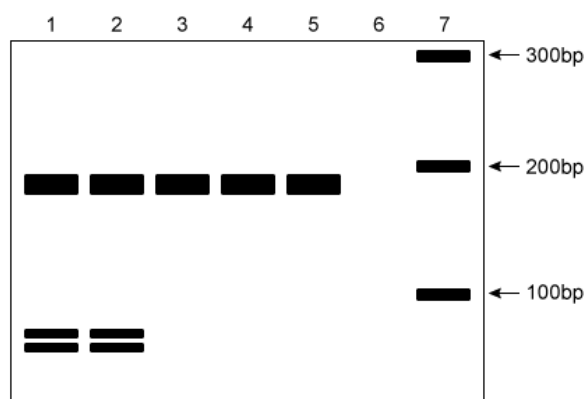


Figure 1 PAGE visualization of PCR product samples following *Apal* digestion. Double helical DNA of known length was used as a ladder (lane 7).

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Gender	Degree of hearing loss	Age at test (years)
Female	None	40
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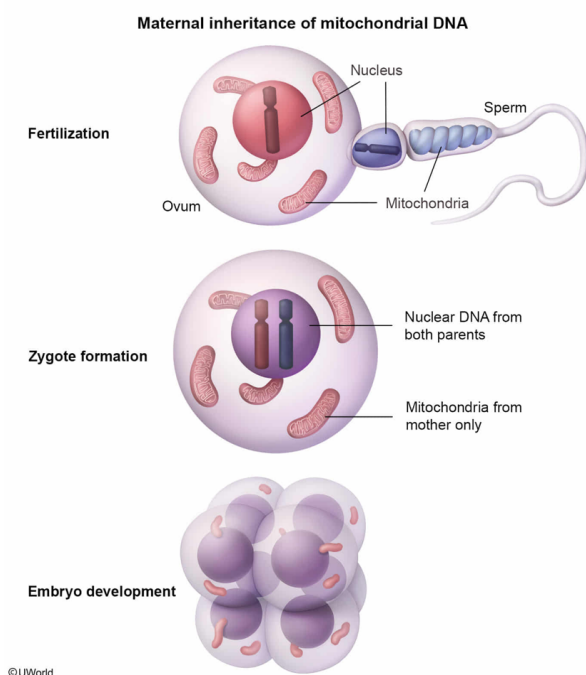
Which of the following is most likely true regarding the inheritance of the A3243G mutation? (Note: Assume affected individuals inherited the mutation.)

- ☐ A. All daughters of an affected father are affected. [11%]
- ✓ ☒ B. Only offspring of affected mothers are affected. [54%]
- ☐ C. All affected males have asymptomatic carrier mothers. [11%]
- ☐ D. Affected offspring must have two copies of the defective gene. [22%]

Correct

54% answered correctly

Explanation:



Mitochondria are cellular organelles that have their own genome; their main functions include ATP generation, heat production, and beta-oxidation of fatty acids.

Mitochondrial genes are extranuclear and inherited in a distinctly different manner than nuclear genes (autosomal and sex-linked). During fertilization, mitochondria within

sperm generally do not pass into the ovum; therefore, typically only maternal mitochondria are transmitted to the fetus. In general, only affected females transmit abnormal mitochondria to offspring; transmission through males is rare, even if they are affected. As a result, the most likely inheritance pattern would be that **only affected mothers** with the **mtDNA A3243G mutation** pass it on to their offspring.

(Choice A) Only one copy of the dominant allele of the gene is necessary to inherit an **X-linked dominant condition**. These diseases are characterized by a lack of father-son transmission, but all daughters of an affected father are affected. Affected mothers have a 50% probability of having an affected child.

(Choice C) Affected males have asymptomatic carrier mothers in **X-linked recessive conditions**, which are inherited via the X chromosome. Two mutant alleles must be inherited to cause disease in females (XX), but only one mutant allele is required to cause the disease in males (XY).

(Choice D) **Autosomal recessive conditions** require the presence of two copies of the defective gene for disease manifestation and show no predilection for males or females. About 25% of children are affected if both parents carry one copy of the defective gene (most common scenario).

Educational objective:

Mitochondria have their own genome, known as mitochondrial DNA, which is generally inherited in a maternal fashion (ie, no paternal contribution). Mitochondria within sperm are not transferred into the ovum during fertilization; therefore, males never pass on their mitochondria.

Time spent:0 sec
Subject:
Biology

QID:400725
System:
Genetics and Evolution

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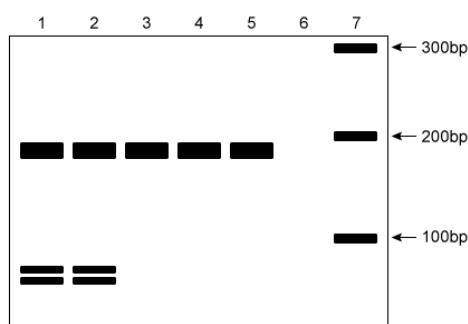


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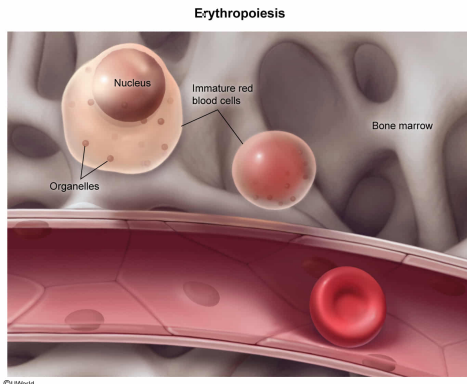
Which of the following cell types would be LEAST sensitive to mitochondrial dysfunction due to mtDNA mutation?

- ☐ A. Hepatocytes [3%]
- ☐ B. Neurons [11%]
- ☐ C. Myocytes [4%]
- ✓ ☒ D. Erythrocytes [79%]

Correct

80% answered correctly

Explanation:



Mature **red blood cells** (RBCs), or erythrocytes, are biconcave, disc-shaped cells that mainly contain hemoglobin, the carrier protein that transports oxygen to body tissues. RBCs are highly efficient transporters of oxygen, **expelling their nucleus and other organelles** during synthesis (erythropoiesis) in the bone marrow to maximize the available space for hemoglobin. As a result, RBCs contain **no mitochondria**; therefore, mitochondrial mutations cannot affect erythrocyte function. RBCs never consume the oxygen they are transporting because they produce energy only via anaerobic glycolysis, an oxygen-independent metabolic pathway that is less efficient than the aerobic one.

Mitochondria allow cells to utilize oxygen in the **electron transport chain** to generate energy in the form of ATP (oxidative phosphorylation), which is why mitochondrial defects tend to primarily affect metabolically active tissues (eg, neural, muscular, heart, kidney). Therefore, the following cells would be affected by mitochondrial dysfunction due to mtDNA mutation:

- Hepatocytes are specialized cells in the liver that contain mitochondria and generate energy for the synthesis of proteins, carbohydrates, and lipids as well as the detoxification of metabolites and drugs **(Choice A)**.
- **Neurons** are specialized nervous system cells that transmit electrical impulses to one another via synapses. Mitochondria in neurons provide energy for the continuous transmission of neuronal signaling **(Choice B)**.
- Myocytes are metabolically active cells within muscles that contain mitochondria, which provide the energy required for muscle contraction and movement **(Choice C)**.

Educational objective:

Erythrocytes are biconcave, disc-shaped cells containing hemoglobin, the carrier protein that delivers oxygen to body tissues. Erythrocytes contain no mitochondria as they expel their organelles during the maturation process in the bone marrow. As a result, mitochondrial mutations do not directly affect erythrocytes.

Time spent: 0 sec
Subject:
Biology

QID: 400726
System:
Genetics and Evolution

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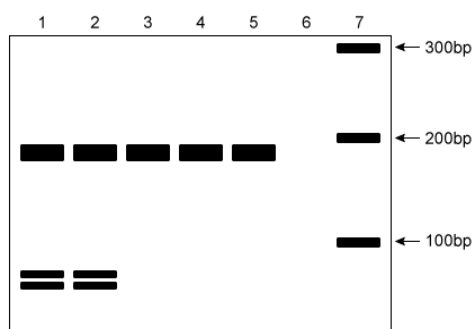


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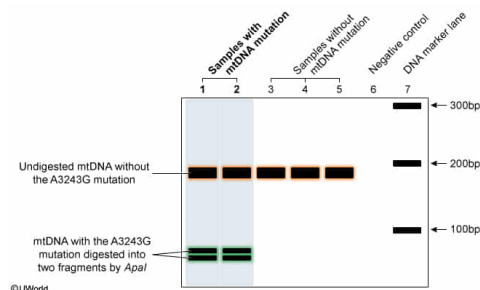
What conclusion can be drawn from the results shown in Figure 1?

- ☐ A. The samples were analyzed under denaturing conditions. [25%]
- ☐ B. The samples were analyzed under reducing conditions. [21%]
- ✓ ☒ C. The samples were analyzed using a native gel. [48%]
- ☐ D. The samples were analyzed using a pH gradient throughout the gel. [4%]

Correct

48% answered correctly

Explanation:



DNA molecules can be separated by size via [gel electrophoresis](#) using a polyacrylamide gel (PAGE) or an agarose gel. The polyacrylamide gel is generally used to analyze proteins or shorter fragments of DNA because it has smaller pores that can slow down the molecules to facilitate their separation. Based on the passage, blood samples were taken from various participants and the mtDNA segment containing the 3243 region was extracted and amplified via PCR. The PCR products were then treated with the restriction enzyme *ApaI* and visualized using PAGE.

Several variations of PAGE exist. Denaturing PAGE uses denaturing agents like sodium dodecyl sulfate (SDS) or urea to denature the molecules being analyzed. For proteins, SDS gets rid of higher levels of protein structure and also imparts a relatively uniform negative charge on the proteins; in this way, proteins all travel toward the anode and separate based on molecular weight alone.

In nucleic acid gels, the sugar-phosphate backbone is already negatively charged, so **no SDS is needed**. Urea may be used to break hydrogen bonds and separate DNA strands, but the ladder in the gel in Figure 1 measures size in base-pairs, indicating that **hydrogen bonding between complementary strands is retained**. Therefore, Figure 1 must show results from a **native gel** that does *not* contain denaturing agents.

(Choice A) In nucleic acid PAGE, a denaturing substance would be added to separate double-stranded DNA into single strands by breaking hydrogen bonds between the two DNA strands. However, Figure 1 indicates that the DNA remains base-paired, with intact hydrogen bonds.

(Choice B) Reducing conditions are used to break disulfide bonds in [protein PAGE](#), *not* DNA PAGE. Because nucleic acids do not form disulfide bonds, no conclusion may be drawn regarding reducing vs. nonreducing conditions.

(Choice D) [Isoelectric focusing](#) uses an immobilized pH gradient throughout the PAGE to separate proteins based on their isoelectric point. As the ionizable groups on DNA all come from the sugar-phosphate backbone, isoelectric focusing is not generally used in nucleic acid PAGE.

Educational objective:

Polyacrylamide gel electrophoresis (PAGE) can be used to separate DNA molecules by size. A native PAGE gel contains no denaturants or reducing agents and allows double-stranded DNA molecules to traverse the gel in their native (unaltered) state.

Time spent: 0 sec

Subject:
Biology

QID: 400727

System:
Genetics and Evolution

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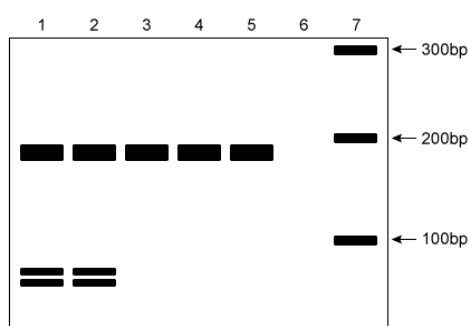


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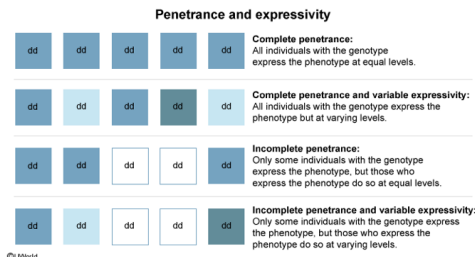
Based on the information given in Table 1, the results of this examination may be explained by all of the following EXCEPT:

- ✓ ☒ A. moderate hearing loss is influenced by autosomal recessivity at the 3243 locus. [35%]
- ☐ B. severe hearing loss is influenced by a single nucleotide mutation. [22%]
- ☐ C. the mutant $tRNA^{leu}$ gene causes variable expressivity of the phenotype. [28%]
- ☐ D. the mutant $tRNA^{leu}$ gene exhibits incomplete penetrance. [14%]

Correct

34% answered correctly

Explanation:



The passage states that family members tested in this study carry no other genetic mutations known to affect hearing loss. Therefore, in these subjects, only the A→G *single nucleotide mutation* at position 3243 in the $tRNA^{leu}$ gene locus of mtDNA influences hearing loss, which ranges from mild to severe (**Choice B**). The results of the audiometry examination, as shown in Table 1, support that this A→G mutation leads to variable expressivity and incomplete penetrance of the hearing phenotype:

- **Variable expressivity** refers to the ability of a single genotype to give rise to multiple phenotypes. In the context of disease, two patients with the same mutated allele may exhibit markedly different clinical symptoms. The A3243G mutation is a prime example of variable expressivity because, as described in the passage, individuals with the mutation exhibit variable phenotypes such as mild to severe degrees of hearing loss (**Choice C**), diabetes mellitus, lactic acidosis, and strokelike episodes.
- **Penetrance** is defined as the proportion of individuals with a genotype who express the corresponding phenotype. When every individual with a specific genotype displays the associated phenotype, the allele is said to show **complete penetrance**. When all individuals have the same genotype but only some express the corresponding phenotype, the condition is said to show **incomplete penetrance** (eg, all patients in Table 1 have the A3243G mtDNA mutation, but not all patients exhibit hearing loss) (**Choice D**).

The question asks which trait does *not* explain the pattern seen in Table 1. **Autosomal** inheritance refers to inheritance of genes encoded by one of the nonsex **nuclear** chromosomes. However, the A3243G mutation discussed in the passage is found on a **mitochondrial** gene. Furthermore, in both male and female offspring, mitochondrial DNA (and any associated mitochondrial disease) is typically inherited only from the mother. Therefore, mitochondrial genes do not follow Mendelian inheritance patterns, which occur when organisms inherit two copies of each gene (ie, one from the father and one from the mother). Due to **heteroplasmy**, the phenotype of a given tissue, organ, or whole organism is influenced by the relative amounts of wild-type and mutant mtDNA, **not by Mendelian patterns** of dominance or recessivity.

Educational objective:

Variable expressivity and incomplete penetrance are common features of mitochondrial diseases. Penetrance is defined as the proportion of individuals with a specific genotype who express the corresponding phenotype. Variable expressivity refers to the range of phenotypic outcomes shown by individuals who carry the same genotype.

Time spent: 0 sec

Subject:
Biology

QID: 400728

System:
Genetics and Evolution

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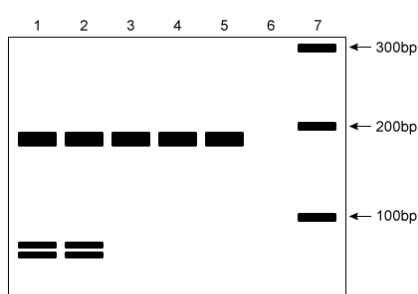


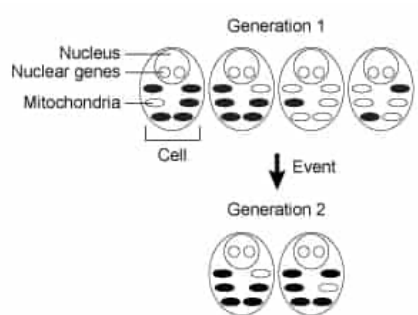
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The image below represents a population of unicellular organisms. The population contains two alleles at a particular mitochondrial gene locus, indicated by shaded and nonshaded mitochondria.



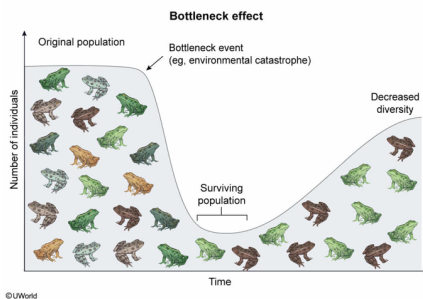
Which of the following events most likely occurred between Generation 1 and Generation 2? Note that the shaded allele is the result of a substitution mutation and is considered to be a deleterious allele in both generations.

- ☐ A. Meiosis [10%]
✓ ☒ B. Population bottleneck [52%]
☐ C. Natural selection [27%]
☐ D. New mutation [9%]

Correct

52% answered correctly

Explanation:



The **bottleneck effect** is the drastic reduction in population size that occurs in response to some sudden and uncontrollable disaster (eg, flood, famine, human-induced catastrophe). In a bottleneck event, no genotype or phenotype is protective, and members of the population are eliminated randomly. This substantially **alters the genetic diversity and allele frequencies** of the population in a way that **does not necessarily follow the principles of natural selection** (where organisms with beneficial alleles are more likely to survive).

For example, an allele can be eliminated completely from the gene pool if no members carrying that allele survive.

Likewise, rare mutant alleles can become more prevalent than wild-type alleles if most members carrying the mutant allele survive and most members carrying the wild-type allele do not. Because **deleterious (ie, harmful)** mutations are most likely selected *against*, which keeps the population of individuals with such mutations low, only large environmental changes like a bottleneck event may allow a **deleterious mutant allele** (eg, in the **shaded mitochondria**) to become more prevalent in the population than the wild-type allele (eg, in the nonshaded mitochondria) (**Choice B**).

(Choice A) Meiosis is a specialized type of cell division that reduces the chromosome number of diploid parent cells by half to generate four genetically distinct haploid gamete cells. If **meiosis** had occurred in this scenario, Generation 1 would have generated 16 haploid cells in Generation 2.

(Choice C) In **natural selection**, alleles that produce environmentally *beneficial* phenotypes are more likely to be selected for and passed on to the next generation than alleles that are harmful. In this scenario, only organisms that primarily express the *deleterious* (ie, harmful) mutant mtDNA survived and reproduced, a result opposite what would be expected had natural selection occurred.

(Choice D) A mutation is described as a permanent, heritable change in the gene sequence. Generation 2 contains the same mitochondrial alleles as Generation 1 (shaded and nonshaded), meaning no new mutation was generated.

Educational objective:

Natural selection is the tendency for alleles that make an organism better suited for survival and reproduction to be passed along to the next generation. Bottleneck events reduce genetic diversity and change the allele frequencies of a population in a random way.

Time spent: 0 sec
Subject:
Biology

QID: 400729
System:
Genetics and Evolution

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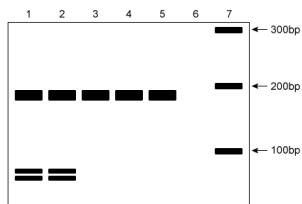


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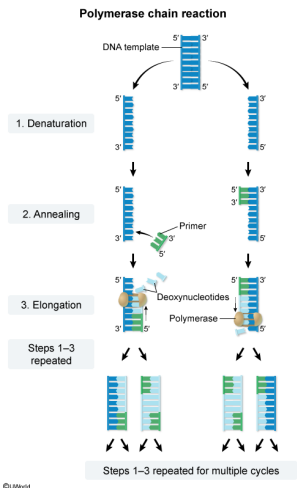
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Based on the passage, which of the following reagents was most likely used during PCR?

- ☐ A. Free ribonucleoside triphosphates [8%]
- ☐ B. RNA polymerase [18%]
- ☒ C. A template sequence containing deoxyribonucleotides [58%]
- ☐ D. Primer pairs with minimal GC content [14%]

Correct
58% answered correctly

Explanation:



Polymerase chain reaction (PCR) uses thermal cycling to amplify small DNA fragments, which can then be screened for mutations or used in further genomic analysis. PCR reagents include the following:

- A **source DNA template** (containing *deoxyribonucleotides*) that includes the target region to be amplified and its adjacent flanking sequences where primers bind
- **Primer pairs** designed from the oligonucleotide sequence of the regions flanking the target sequence
- A **thermostable DNA polymerase** (ie, not denatured at high temperatures) to replicate the DNA template using a pool of supplied deoxyribonucleoside triphosphates (dNTPs)
- A **buffer solution with positively charged ions** (cations) to provide an optimal environment for DNA polymerase to function (ie, cations bind the negatively charged phosphates on the DNA backbone and those on dNTPs, neutralizing the negative charge of DNA and stabilizing primer-template binding)

In PCR, thermal separation (denaturation) of the DNA template is accomplished by exposing the sample to high temperatures. Subsequent cooling allows primers to

anneal to the single-stranded flanking ends of the target region. DNA polymerase uses the primers to elongate the new daughter strands in the 5' to 3' direction. These steps are repeated to obtain millions of copies of the target DNA segment in a short time. The passage states that the mitochondrial (DNA) genome was amplified by PCR; therefore, a **template sequence containing deoxyribonucleotides** was used during the PCR procedure.

(Choice A) Deoxyribonucleoside triphosphates (dNTPs), not ribonucleoside triphosphates, are used during PCR. Because the PCR product is DNA (*not RNA*),

deoxynucleotides that have a deoxyribose sugar (ie, deoxygenated on the 2' carbon) must be used. In contrast, ribonucleoside triphosphates (NTPs) contain *ribose* and are used during transcription to produce RNA.

(Choice B) PCR produces DNA products. RNA polymerase is not used during PCR because it synthesizes new RNA molecules from DNA during transcription.

(Choice D) GC-rich DNA fragments tend to be more stable at high temperatures; therefore, primers must have a *substantial amount of GC content* (40%-60%) to avoid premature denaturation during PCR.

Educational objective:

Polymerase chain reaction (PCR) is a thermal cycling technique used to amplify DNA fragments. PCR reagents include a source DNA template, GC-rich primer pairs, a thermostable DNA polymerase, and a buffer solution with positively charged ions.

Time spent: 0 sec
Subject:
Biology

QID: 400730
System:
Genetics and Evolution